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Design and Evaluation of Blockchain-Enabled Payment System Using API Integration for Enhanced User Experience

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Abstract— *This paper proposes PayAPIChain, a blockchain-based institutional payment system that links smart contracts with RESTful APIs to support university fee transactions with transparency, traceability, and auditability. Deployed on the Ethereum-compatible Sepolia test network, the prototype provides authenticated API calls for payment initiation, request validation, status tracking, and confirmation, while storing personally identifiable information (PII) off-chain to reduce on-chain exposure and comply with privacy requirements. System quality was evaluated using seven ISO/IEC 25010 software characteristics together with usability feedback from 40 participants (5 ICT experts and 35 graduate students). The evaluation yielded weighted mean scores from 3.53 to 3.87, interpreted as "Very Acceptable." In performance trials across representative university fee scenarios, the system achieved an average end-to-end transaction confirmation time of 1.2 s ± 0.3 and an API success rate of 98.6%. Unlike blockchain payment approaches that rely on direct end-user wallet interaction, PayAPIChain is designed for tighter integration with institutional web applications through an API-mediated workflow, enabling centralized policy enforcement, consistent audit logs, and easier adoption for non-technical users. The results indicate that API-based blockchain payment systems can be feasible for higher education institutions seeking secure and user-friendly digital payment options, provided that key management, access control, and monitoring are implemented as operational controls. The prototype was validated on a test network, and future work includes pilot deployment and stress testing under peak enrollment loads.*

Keywords— blockchain; rest api; iso/iec 25010; online payment system; higher education; institutional payments; data privacy; usability evaluation

I. INTRODUCTION

Blockchain is a secure, decentralized and immutable transaction recording structure [10] [14]. However, traditional Web3 dApps expose low-level (wallet/ smart contract) interfaces that are complex for non-technical users and do not fit well with the existing organizational processes [11]. Introducing blockchain via stateless RESTful APIs could help to maintain transparency and auditability whereas making the system more user friendly, as well as increase compatibility with existing university systems [11], [12], [15].

At higher-education, blockchain has been trialled for administration and governance of the academic system and in the verification of † credentials [1]–[3],[9]. However, the majority of the prototypes are purely focused either on credentialing or they factor in end-users who need to handle wallets directly which restricts them from being used in actual institutional deployment scenarios. PayAPIChain fills this gap thereby, brokering all of the operations on Ledger Using REST API layer which can be plugged into university's payment portals.

However, despite the recent digitization attempts, most of the higher education payment systems still face uncomfortable delay in posting transactions, lack transparentness and vulnerabilty to manual mistakes or fraud [2], [3]. The REST API-based blockchain payment applications provide secure and traceable and near-real-time settlement, while enabling agile iterative development as well as systematic quality assessment based on ISO/IEC 25010 [4], [5], [13]. In the Philippine setting, efforts like the PHPC peso-backed stablecoin indicate a burgeoning appetite for regulated blockchain-based financial architecture [8].

This paper presents the design and implementation of PayAPIChain, an API-driven blockchain payment prototype that is adapted with the institutionalized payment scheme at Nueva Ecija University of Science and Technology (NEUST). Specific objectives of the project are as follows: (1) to develop an API-based payment framework that traces institutional transactions onto a tamper-evident blockchain ledger while preserving Personally Identifiable Information (PII) off-chain, (2) to deploy and release said solution on an Ethereum-compatible test net hive under real-like scenarios for institutional payments, and (3) to evaluate system's quality regarding functional suitability, usability, portability and overall acceptability in accordance with ISO/IEC 25010:2011 standard on software product quality.

With these objectives in mind, the study investigates the following research questions:

RQ1. How can one design an API-based payment Blockchain architecture for NEUST to record institutional transactions on a tamper proof ledger and store PII off-chain?

RQ2. What are the transaction-confirmation times, and issue-operation success rates, of the PayAPIChain prototype under typical institutional payment cases on an Ethereum-compatible test network?

RQ3. What are the IT professionals and graduate students' perceptions with respect to functional suitability, usability, portability, and overall acceptability of PayAPIChain as per ISO/IEC 25010:2011?

A. Novelty and Contributions

Distinct from existing works of HEI blockchain payment that just concentrate on credentialing, or need learners to use wallet/dApp directly [1]–[3], [9]–[12], [15], PayAPIChain integrates the properties of usability and integration into the touchpoints consuming REST API in NEUST's web payment flows to facilitate ease of use and take full advantage of users' familiar with them while maintaining on-chain transparency and auditability.

B. Our Contributions

This study: (1) unfolds an API-mediated Web UI → REST API → smart contract system design for HEI payments on an Ethereum-compatible network; (2) presents ISO/IEC 25010 ratings from 5 IT professionals and 35 graduate students on functional suitability, usability, portability and global acceptability;(3) showcases a privacy-preserving deployment allowing PII to remain off-chain manner while providing token-based authentication and audit logging for institutional governance [1], [2], [4], [10], [11], [21], [25], [26].

II. RELATED WORK

Blockchain in education has been widely investigated for its application to improve transparency, traceability and administrative effectiveness regarding credentialing, records management and governance [1]–[3]. The existing literature [1], [2] recommended the application of blockchain technologies in real universities and not only as theoretical demonstrators. In the field of payment systems, Toshniwal and Jaiswal show how real-time transparent payments can be made through blockchain technology. Furthermore, BSP's PHPC that is pegged to peso can be a model of substitute to tokenized fiat for compliant, peso-denominated tuition and fee settlements in the Philippines [8], [9], [12]. However, these approaches generally still require that end-users wallet or smart contract interaction.

More and more industry and design studies propose API-based integrations, where blockchain backends are used behind HTTP web APIs. This way of doing things keeps the familiar practice while allowing for ledger-level auditability [4, 6, 7, 21]. These architectures can be evaluated through the ISO/IEC 25010 quality model (functionality, usability and portability among others). It will enable more standardised reporting of software quality within blockchain-enabled HEI systems at the same time as it will help to establish hook-ups between PHPC-style infrastructures and school payment platforms [4], [8].

A. Contrast with our work

The majority of blockchain projects in education focusing on credentialing and record-keeping, or even proposing abstract payment schemes that rarely provide an API layer with user and expert evaluation based on ISO/IEC 25010. By contrast, in this work we employ a REST-based payment scheme (web UI → REST API → smart contract) compatible with institutional practices. It is built-in with privacy-preserving off-chain for PII and collected feedback from 5 experts and 35 students on selected ISO/IEC 25010 quality characteristics [1]–[3], [7], [9], [21]. Table I compares the representative blockchain-based education and payment systems in terms of their focus, interface style, evaluation lens and the HEI payments they are applicable to.

TABLE I. CONTRAST TABLE

Work	Education focus	Interface style	Evaluation lens	Applicability to HEI payments
Raimundo & Rosário 2021 [1]	Broad education uses (transparency, governance)	Mixed (conceptual)	Narrative/landscape	Indirect (not payment-specific). (ScienceDirect)
Bhaskar et al. 2020 [2]	Univ. integration & social impact	Integration concepts	Social impact	Indirect (institutional context). (MDPI)
Toshniwal & Jaiswal 2023 [9]	HEI registration fees	Wallet/DApp oriented	Design-centric	Direct to fee collection; limited API/UX evaluation. (CEUR-WS)
Gomez 2023 [21]	IS-module integration	API-mediated (programmatic)	Case/architecture	Strong for integration; not HEI-specific. (ScienceDirect)

Blockchain-Enabled Payment System Using API Integration for Enhanced User Experience	HEI payments (NEUST)	REST API → smart contract	ISO/IEC 25010; users + experts	Direct: evaluated, privacy-aligned, deployable.

III. DATA AND METHODS

The researchers employed a descriptive-developmental framework alongside an Agile interpretation of the Software Development Life Cycle (SDLC), which consists of planning, analysis, design, implemenartion and testing: to develop a blockchain-based online payment system with RESTful API for NEUST. Transactions—featuring pseudonymous identifiers, institutional references, test Ether (ETH) amounts, and timestamps—are registered on an immutable transparent ledger; a RESTful interface connects transactions with end-user interactions [10], [11], [14], [15], [17]. Security and usability followed the benchmarks established for financial blockchains, while functional modules were tactically reviewed by IT professionals and final users to verify adherence with the requirements as well as ISO/IEC 25010 software quality model [4], [6], [12], [15],[22],[23].

A. Implementation Details: REST API, Authentication Flow, and Smart Contract Interface

This section summarizes the REST API endpoints, authentication flow, and core smart contract of the PayAPIChain prototype so that other institutions can reproduce the architecture without handling private keys or raw blockchain RPC calls directly.

1) REST API specification

The PayAPIChain middleware acts as an institutional façade that wraps low-level blockchain operations behind a JSON-based HTTPS API. Endpoints accept POST requests with an institutional API key in the header and return JSON responses containing a Success flag, a human-readable Description, and an optional payload.

Table II lists the main REST endpoints, their HTTP methods, required request fields, and key response fields. All requests follow a common JSON pattern. For example, wallet creation via /adduser can be invoked as:

```
{
 "privatekey": "<PRIVATE_API_KEY>",
 "studentno": "1001"
}
```

with a typical response:

```
{
 "Success": true,
 "Description": "Student's wallet has been created",
 "Wallet": "0x1234...abcd",
 "Balance": 0,
 "ProgrammerID": 2
}
```

Other endpoints (e.g., balance queries and payment submission) reuse the same structure, differing only in their specific fields and functions as summarized in Table II.

TABLE II. CORE PAYAPICHAIN REST ENDPOINTS USED IN THE PROTOTYPE

Endpoint	Method	Purpose	Required fields (request body)	Key response fields (success)
/adduser	POST	Create or attach a blockchain wallet to a student number.	privatekey, studentno	Success, Description, Wallet, Balance, ProgrammerID
/balanceeth	POST	Retrieve the ETH balance for a student's wallet.	privatekey, studentno	Success, Description, Wallet, Balance
/balancepeso	POST	Retrieve the EPhp token balance for a student's wallet.	privatekey, studentno	Success, Description, Wallet, Balance
/sendpesoto	POST	Send EPhp tokens from a student wallet to an institutional wallet.	privatekey, studentno, value (amount in EPhp), to (destination address)	Success, Description, Listarray (includes tx hash)
/listofusers	POST	List all student numbers registered under the institutional API key.	privatekey	Success, Description, list of studentno
/transtatus/{hash}	GET	Query the status of a blockchain transaction by its hash.	– (hash in URL path)	Success, Description, TransactionHash, Status
/faucet/{wallet} *	GET	(Test only) Request a small amount of EPhp for a wallet.	privatekey (in JSON body)	Success, Description, optional updated Balance

2) Authentication and backend integration

Authentication follows a two-tier flow. Students first log in to the NEUST web application via institutional SSO and, once a session is issued, access a payment dashboard showing their current and outstanding balances. The NEUST back end then authenticates to the PayAPIChain service with a server-side API key and acts as a trusted proxy, so all PayAPIChain requests originate from the institutional server and no API credentials are exposed in the browser. In this setup the back end creates and stores student wallets on first login, performs balance queries and payment submissions through the REST endpoints, and verifies that each on-chain transaction has reached "Confirmed" status before updating local payment records and reconciliation logs.

3) Smart contract interface and ABI fragments

Internally, PayAPIChain interacts with two on-chain assets: (i) the native ETH coin, used to pay transaction fees, and (ii) an EPhp token implemented as an ERC-20–style smart contract that represents a peso-pegged unit of value. The EPhp contract exposes only the minimal interface needed for the prototype, namely the balanceOf and transfer functions and the standard Transfer event:

```
function balanceOf(address account) external view returns (uint256);
function transfer(address to, uint256 value) external returns (bool);
event Transfer(address indexed from, address indexed to, uint256 value);
```

Any EVM-compatible network that deploys a contract with this interface and is reachable from the API middleware can be used to reproduce the experiments reported in this paper.

4) Sequence diagram for a typical institutional payment

The overall system architecture and data flow are summarized in Figs. 1 and 2, while Fig. 3 presents the message sequence among the Student (web browser), NEUST Web Application (PHP backend and database), the PayAPIChain REST API, the Sepolia blockchain (ETH and EPhp contracts), and the institutional database. A typical tuition or document-payment workflow is: (1) the student logs in and the web app loads outstanding balances; (2) the backend ensures a wallet exists via /adduser and stores the address; (3) the dashboard requests balances through /balanceeth and /balancepesoto; (4) when the student proceeds to pay, the backend calls /sendpesoto to trigger an on-chain EPhp transfer; (5) the API returns a transaction hash, which is stored with the payment record; and (6) the web app polls for status and marks the institutional record as paid once confirmed. Together with Fig. 3, this sequence clarifies the separation between on-chain and off-chain components and can be adapted to other institutional environments [11], [19], [21].

5) Security considerations: threat model, mitigations, and validation tests

In addition to blockchain vulnerabilities, PayAPIChain's engineering web applications also carry significant risks. Key threats include pilfering student credentials, client-server communication modifications, misuse of the apikey and EPhp token contract abuses. Mitigations include using an institutional SSO with role-based access control, enforcing HTTPS/TLS, keeping API and signing keys on one's own server, and limiting EPhp contract operations by the other parameters of a policy-protected team. The prototype was subjected to S1–S3 test scenarios with regard to input validation, bypass authentication attempts, replays, and automated scans. No high-severity issues were found, though formal penetration testing and hardening must precede production.

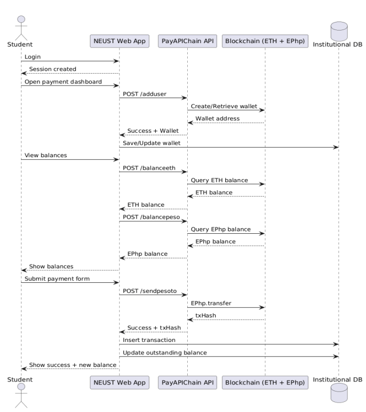

Fig. 1. Sequence diagram of a blockchain-mediated institutional payment transaction using the PayAPIChain REST API and EPhp token transfers

6) Deployment and operational considerations

The contract owner's key for EPHA and the hot wallet of the institution should be managed jointly between finance and IT departments, with only a small number of trustees allowed to participate. Secure backup and key rotation procedures are vital, as are clearly prescribed protocols regarding who may update contracts or swap keys, who can validate significant transfers and how to handle suspected key breaches. While the experiments used the Sepolia test network, this architecture could also be deployed on the Ethereum mainnet or a peso-backed stable coin. Early estimates indicate that gas costs for tuition-sized payments remain reasonable, subject to ongoing attention to fee monitoring and the general state of the network [24]. By recording the transaction hash, amount and time of each successful payment, reconciliation is made simpler. Each step here is checked off by cross-cutting between an institutional database on the one hand and on-chain transfers on the other. This makes it easy for you to do internal accounting as well as for free external auditing.

TABLE III. BLOCKCHAIN EXPERIMENTAL SCENARIOS FOR PAYAPICHAIN

Scenario	Institutional payment context	Web/API action	Blockchain interaction	Expected outcome
S1 – Registration fee	Student pays regular registration or enrollment fees for a term.	Student completes the registration fee form in the web interface; REST API receives and validates the request.	REST API calls the smart contract on the Sepolia test network to record a registration-fee transaction (student identifier, institutional reference number, amount in test ETH, timestamp).	A confirmed Sepolia transaction hash is returned; the system marks the registration fee as "Paid" and the payment record becomes visible to both student and registrar.
S2 – Document-processing fee	Student requests official documents (e.g., transcript of records, certifications).	Student submits a request specifying document type and associated fee; REST API verifies the amount and request details.	Smart contract writes a document-fee transaction linked to the document request ID to the blockchain ledger.	The transaction hash is stored; the document request status is updated to "Payment verified," allowing staff to process the document.
S3 – Additional / miscellaneous fees	Student settles penalties, surcharges, or other miscellaneous institutional fees.	Student or staff encodes the fee details via the web interface; REST API receives and validates the request.	Smart contract logs an additional-fee transaction associated with the student's institutional account on Sepolia.	A transaction hash is generated and persisted; the corresponding miscellaneous fee record is updated as "Cleared" in the institutional system.

In a series of web/post with the tag "Digital Identity Awareness," this research evaluates PayAPIChain using selected ISO/IEC 25010:2011 characteristics (functional compatibility, usability, portability) and overall acceptability measurements.A 4-point Likert survey of 5 ICT experts and 35 graduate students (N = 40) produced highly repeatable results (Cronbach's α=0.91) [4], [28], [29].End-to-end tests of the Web UI REST API smart contract pipeline on the Sepolia network turned up ≈1.2 s average duration from request to approval, a 98.6% API success rate, and tuition-sized gas costs all the means fall within reasonable limits. Table IV scores all "Very Acceptable," indicating readiness for HEI pilot deployment with the EPhp smart contract as a trustable ledger and student records maintained on institutional systems [7], [11], [14], [17], [19],[20],[21].

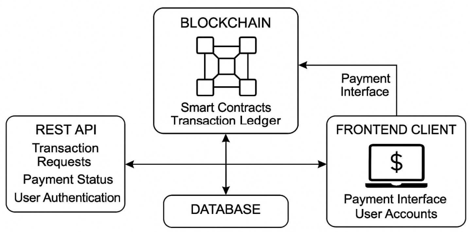

Fig. 2. System architecture of the blockchain-enabled online payment system using REST API integration.

Fig. 2 shows the process flow of the blockchain-enabled online payment system, from the student's interaction with the NEUST web application through the forwarding of validated requests to the PayAPIChain REST API and the recording of transactions on the blockchain ledger with feedback to the user. It highlights key checkpoints such as input validation, transaction submission, smart contract execution, and confirmation of the payment state.

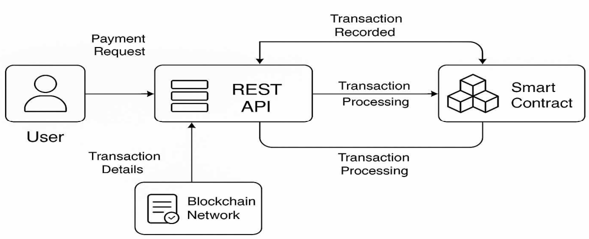

Fig. 3. Process flow of the blockchain-enabled online payment system via REST API integration.

Table III shows that the API-based blockchain payment ecosystem obtained an overall mean of 3.77, with functional suitability (3.79), portability (3.78), overall acceptability (3.77), and usability (3.74) all falling within the "Very Acceptable" band defined in Table IV. Combined with an average end-to-end verification time of about 1.2 s and solid cross-platform performance on desktop, laptop, and mobile devices, these results suggest that students and staff can submit and confirm payments quickly and with minimal friction, representing a clear improvement over manual workflows and earlier, less integrated blockchain prototypes [21].

TABLE IV. SUMMARY OF ISO/IEC 25010 EVALUATION OF PAYAPICHAIN

Evaluation Criteria	N	Weighted Mean	Interpretation
Functional Suitability	40	3.79	Very Acceptable
Usability	40	3.74	Very Acceptable
Portability	40	3.78	Very Acceptable
Acceptability	40	3.77	Very Acceptable
Overall Mean	40	**3.77**	**Very Acceptable**

Note: The weighted mean scores in Table III are interpreted using the four-bar scale which accords with 4-point response options (1=Not Acceptable, 2 = Less Acceptable, so on and so forth). Prior Philippine evaluation studies have been based on factors of this kind.

Functional suitability received the highest mean score, at 3.79. This confirms that core payment characteristics function well under typical use there by users who once initiated and checked out transaction data; additionally, portability (M.3738=Very Acceptable) is compatible in different environments while maintaining crossbreed adapters.

Considering that the first three are all Very Acceptable, the conclusion is that the system is indeed suitable for limited tests in institutional payment transactions, even though it has yet to win approval as a mature large-scale development product.

TABLE V. INTERPRETATION SCALE FOR WEIGHTED MEAN

Weighted Mean Range	Interpretation
3.50 – 4.00	Very Acceptable
2.50 – 3.49	Acceptable
1.50 – 2.49	Less Acceptable
1.00 – 1.49	Not Acceptable

These results suggest that the proposed system is a viable candidate for blockchain-based payment deployment in higher-education institutions seeking to modernize financial management, provided that additional technical, organizational, and regulatory checks are addressed. Respondents also highlighted the value of reduced face-to-face contact and shorter queues, noting that the system lessens manual interaction and streamlines payment processing. Taken together, these findings align with the broader move toward digital trust systems that give users more control over their financial transactions and are consistent with the original vision of decentralized, trust-minimized payments in blockchain systems [18].

IV. CONCLUSION

This study implemented and evaluated PayAPIChain, a blockchain-enabled online payment system that integrates an Ethereum-compatible smart contract with a REST API and NEUST's institutional payment workflows. By hiding low-level wallet operations behind an API-mediated web interface while retaining PII in institutional systems, the prototype improves transparency, security, and user experience for routine HEI payments. ISO/IEC 25010 results from 5 experts and 35 graduate students show "Very Acceptable" ratings for functional suitability, usability, portability, and overall acceptability, indicating that the architecture is a viable candidate for institutional pilots.

A. Limitations

The evaluation used scripted test transactions on the Sepolia test network and was conducted in a controlled environment, so scalability and robustness under real high-volume conditions (e.g., peak enrollment periods) still need to be validated [24]. The study also focused on one institution and a limited set of payment scenarios.

B. Future Work

Future work will involve live pilots in partner HEIs, multi-institutional benchmarking of performance and user adoption, and integration with government-endorsed stablecoins, national QR-code standards, and mobile e-wallets [8], [16], [24]. Additional enhancements will explore automated handling of scholarships and tuition instalment plans while keeping gas costs and service levels within acceptable bounds. Beyond technical improvements, the design aligns with Environmental, Social, and Governance (ESG) principles and UN Sustainable Development Goal 9 by supporting cashless payments, reducing paper use, and fostering transparent, trust-oriented digital financial services in education [10], [27].

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APPENDIX
SURVEY INSTRUMENT AND TEST SCRIPT SUMMARY

This appendix provides a concise summary of the survey instrument and the API test configuration used to evaluate the PayAPIChain prototype. The complete expert and student questionnaires and full test scripts are available from the authors upon request.

A. Survey Instrument (Expert and Student Versions)

Both expert and student respondents rated the system using ISO/IEC 25010-based items for functional suitability, usability, portability, and overall acceptability. Table A.1 shows sample items from the instrument.

TABLE A.1 SAMPLE ITEMS FROM ISO/IEC 25010-BASED SURVEY INSTRUMENT

Construct	Sample item for students	Sample item for experts
Functional suitability	"The system correctly processes my institutional payments."	"The system's functions are adequate for typical HEI payment workflows."
Usability	"The interface is easy for me to learn and use."	"The user interface supports efficient completion of payment tasks."
Portability	"The system works properly on the devices I commonly use."	"The system can be deployed on different platforms with minimal changes."
Overall acceptability	"Overall, I am satisfied with using this system for payments."	"Overall, the system is acceptable for institutional pilot deployment."

All items used a 4-point Likert scale: 1 = Not Acceptable, 2 = Less Acceptable, 3 = Acceptable, 4 = Very Acceptable.

B. API Test Script Configuration (Summary)

To validate the REST API and blockchain integration, an automated script exercised the Sepolia deployment by loading the API base URL, institutional API key, and test wallets, then calling /adduser, /balanceeth, and /balancepeso to verify wallet creation and balances, /sendpesoto to submit a fixed test payment, and /transtatus/{hash} to poll for confirmation and log time, gas usage, and errors. These steps were repeated for the three experimental scenarios (registration, document-processing, and miscellaneous fees) to derive the average confirmation times and API success rates reported in the main text.